

GALAXY

LIMITER

User manual • Version 1.1

Mastering limiter with 10 limiting styles • VST3 • AU • Standalone

Contents

1	Welcome
2	Installing and building
3	Quick start
4	The interface
5	Controls reference
6	Style, oversampling, true peak and dither
7	The 10 limiting styles
8	The built-in clipper
9	Monitoring: Unity gain and Delta
10	Meters and loudness
11	Latency
12	Troubleshooting
13	Specifications
14	Backup contents and version history

1. Welcome

Galaxy Limiter is a mastering limiter for the end of your mix or master chain. It raises loudness while making sure your audio never goes above the output ceiling you set. What makes it different is the **Style** menu: ten limiting styles, each with its own attack shape, release behaviour, knee, clipping and harmonic colour, from completely clean to loud and gritty.

Main features

- Lookahead brickwall limiter with a guaranteed output ceiling
- 10 limiting styles, with an integrated clipper and optional analog-style colour
- Oversampling from 2x to 16x, and a true-peak mode for streaming-safe masters
- Unity-gain and Delta monitoring to judge what the limiter is really doing
- Scrolling level and gain-reduction display, peak meters and EBU R128 loudness (LUFS)
- TPDF dither to 16 or 24 bit
- VST3, AU and standalone app. Works in Logic Pro, Cubase, LUNA, Fender Studio Pro and other hosts

2. Installing and building

Galaxy Limiter is delivered as source code, so you build it once on your computer. The build downloads the JUCE framework automatically, so you need an internet connection the first time.

What you need on a Mac

- The Xcode command line tools: run `xcode-select --install` in Terminal.
- CMake: install Homebrew, then run `brew install cmake`.

Building for the first time

Put the `GalaxyLimiter` folder in Downloads, open Terminal and paste these commands one block at a time:

```
cd ~/Downloads/GalaxyLimiter
rm -rf build
cmake -B build -DCMAKE_BUILD_TYPE=Release
```

```
cmake --build build -j8
```

```
killall -9 AudioComponentRegistrar
```

The build takes about 5 to 10 minutes the first time. The plugins are copied automatically to:

Format	Location
VST3	~/Library/Audio/Plug-Ins/VST3/Galaxy Limiter.vst3
AU	~/Library/Audio/Plug-Ins/Components/Galaxy Limiter.component
Standalone	GalaxyLimiter/build/GalaxyLimiter_artefacts/Release/Standalone

Rebuilding after changes

After replacing files in the Source folder, you only need the build step. It is much faster than the first build:

```
cd ~/Downloads/GalaxyLimiter
cmake --build build -j8
killall -9 AudioComponentRegistrar
```

Then quit and reopen your DAW.

> **Tip:** never paste the contents of a script file straight into Terminal. Paste the commands shown here instead. If a line starts with `#!`, zsh will show “event not found”.

Windows

Install Visual Studio 2022 (with “Desktop development with C++”), CMake and Git for Windows. In Git Bash, run as administrator:

```
cd ~/Downloads/GalaxyLimiter
cmake -B build -G "Visual Studio 17 2022" -A x64
cmake --build build --config Release
```

Windows builds VST3 and the standalone app. AU only exists on macOS.

3. Quick start

A simple routine that works for most masters:

1. Put Galaxy Limiter as the **last plugin** on your master or stereo bus.
2. Set **Output ceiling** to -1.0 dB and turn **True peak** on. This is safe for streaming services.
3. Choose a style. Start with **Transparent** or **Modern**.
4. Raise **Gain** slowly while watching the pink gain-reduction meter. 2 to 4 dB of reduction on the loudest parts is a common, safe amount.
5. Turn on **Unity gain** and switch it on and off. This compares the tone without the loudness change, so you hear whether limiting is hurting the mix.
6. Try a few styles. With Unity gain on, the comparison is fair.
7. Check **Integrated** loudness while playing the whole song. Press **Reset loudness** before each full pass.
8. Turn **Delta** off before you export.

Typical loudness references

Most streaming services turn loud songs down to their own playback level, so extreme loudness rarely helps. These are common reference points, not rules:

Use	Integrated loudness	Ceiling
Streaming (general)	around -14 LUFS	-1.0 dB, true peak on
Loud modern pop, EDM, hip-hop	-10 to -7 LUFS	-1.0 dB, true peak on
Dynamic genres (jazz, classical, acoustic)	-18 to -14 LUFS	-1.0 dB
CD / WAV master	to taste	-0.3 dB, dither 16-bit

4. The interface

The window is divided into four areas.

Area	What it contains
Top bar	Logo, Style menu, Oversampling menu, True peak switch and Dither menu.
Display (centre)	A scrolling history of about 8 seconds. Violet shows the input level, cyan shows the output level, and the pink line hanging from the top shows gain reduction. The dashed magenta line is the output ceiling. The text below the display describes the current style.
Meters (right)	Output peak meters for left and right, the gain-reduction meter (GR), and loudness readouts: Momentary, Short-term and Integrated LUFS. Reset loudness is below them.
Control panel (bottom)	Six knobs: Gain, Output ceiling, Attack, Release, Lookahead and Stereo link. On the right are the Unity gain and Delta switches.

Using the knobs

- Drag up or down to turn a knob.
- Double-click the value under a knob to type an exact number.
- All controls can be automated from your DAW and are saved with your session.

5. Controls reference

Control	Range	Default	What it does
Gain	0 to +30 dB	0 dB	How hard you drive the limiter. More gain means more loudness and more gain reduction.
Output ceiling	-12 to 0 dB	-1.0 dB	The maximum output level. Nothing goes above it.
Attack	0.01 to 10 ms	2 ms	How gradually gain reduction fades in before a peak. Longer is smoother and cleaner; shorter keeps more punch. It can never be longer than Lookahead.
Release	1 to 2000 ms	100 ms	How quickly the limiter recovers after a peak. Short is louder but can pump or distort; long is smoother but less loud. Each style also adds its own program-dependent release.
Lookahead	0 to 10 ms	3 ms	How far ahead the limiter looks, so it can turn down before a peak arrives. More lookahead is cleaner but adds latency.
Stereo link	0 to 100 %	100 %	100 % reduces both channels equally, keeping the stereo image stable. Lower values let each side limit on its own, for more loudness but a slightly wandering image.
Unity gain	on / off	off	Monitoring only. Removes the Gain boost so you can compare tone at equal loudness. See section 9.
Delta	on / off	off	Monitoring only. Plays only what the limiter removes. See section 9.

6. Style, oversampling, true peak and dither

Style

Chooses one of the ten limiting styles described in section 7. You can switch styles while playing.

Oversampling

Runs the limiter at a higher sample rate internally (2x, 4x, 8x or 16x). This reduces aliasing, a harsh distortion that fast limiting and clipping can create, and catches peaks between samples more accurately. Higher settings sound cleaner but use more CPU and add a little latency.

Setting	When to use it
Off	Tracking, low CPU, or when the limiter is barely working
2x or 4x	Everyday mixing and most masters
8x or 16x	Final export, heavy limiting, or clipping styles (Aggressive, Punchy, FET Analog)

True peak

When digital audio is converted to analog or to MP3/AAC, the real waveform can briefly go higher than the sample values show. These are called inter-sample or true peaks. True peak mode uses at least 4x oversampling and keeps an extra 0.1 dB of safety so your master stays under the ceiling after conversion. Leave it on for anything going to streaming services.

Dither

Adds a tiny amount of noise before the audio is reduced to a lower bit depth, which prevents a grainy kind of distortion on quiet passages. Use it only on the **final** export, and only once in the chain.

Setting	Use it when
Off	Exporting 32-bit float, or a later processor will dither
16-bit	Exporting a 16-bit WAV, for example for CD
24-bit	Exporting a 24-bit WAV

7. The 10 limiting styles

All ten styles use the same core limiter. They differ in how they shape the gain reduction and how much they use the clipper and harmonic colour. With Unity gain on, you can compare them fairly.

Style	Sound and best use
Transparent	Clean brickwall with smooth attack and program-dependent release. The lowest distortion. Good first choice for any material.
Punchy	Shorter attack; lets transients poke through into a soft clipper for extra snap. Drums, rock, pop.
Dynamic	Strong program-dependent release that keeps groove and micro-dynamics. Funk, acoustic, anything that must still breathe.
Aggressive	Fast and loud, with heavy transient clipping. EDM, trap and loud modern genres. Watch for distortion.
Modern	Balanced two-stage release for loud but clean masters. A great all-rounder.
Safe	Slow and careful. Avoids distortion even with heavy gain reduction. Orchestral, jazz, podcasts.
Bus Glue	Wide soft knee and slow release; behaves like a gentle bus compressor going into a wall. Gluing a mix together.
Vari-Mu Tube	Wide knee, slow program release and warm 2nd-harmonic colour. Adds thickness and warmth.
FET Analog	Very fast attack with an edgy odd-harmonic bite, like a pushed FET limiter. Adds aggression and presence.
Opto	Smooth optical-style release that gets slower the harder you push it. Vocals-heavy mixes, soft and musical limiting.

The colour styles (Vari-Mu Tube, FET Analog, Opto and, lightly, Bus Glue, Aggressive, Punchy and Modern) only add harmonics while the limiter is actually reducing gain, like real analog hardware. With no gain reduction they pass the audio unchanged.

8. The built-in clipper

Galaxy Limiter includes a clipper in its final stage. There is no separate knob for it: each style sets how it behaves. It works in three layers.

1. Style-controlled output clipper

After the limiter, every sample passes through a clipper set at your ceiling, so nothing can go above it. Each style decides how soft that clipper is. A soft clipper rounds peaks gently and starts working up to 3 dB below the ceiling; a hard clipper only acts right at the ceiling.

2. Transients sent into the clipper

Some styles deliberately let the loudest, shortest transients skip the limiter and hit the clipper instead. Clipping a very short drum hit often sounds punchier and louder than turning it down.

Style	Clipper shape	Transients sent to clipper
Safe	Hard (emergency catch only)	none
Transparent	Almost hard	none
Modern	Medium soft	0.8 dB
Dynamic	Medium soft	0.5 dB
Bus Glue	Soft	0.3 dB
Opto	Soft	0.3 dB
Punchy	Soft	1.5 dB
Vari-Mu Tube	Very soft	0.5 dB
FET Analog	Very soft	2.0 dB
Aggressive	Very soft	3.0 dB

3. Final safety clip

After oversampling, a last hard clip at the ceiling guarantees the exact sample-peak level. It almost never does anything audible.

Tip: clipping creates harmonics, which can turn into aliasing. When using Aggressive, Punchy or FET Analog, set Oversampling to 4x or higher.

9. Monitoring: Unity gain and Delta

Unity gain

Louder almost always sounds “better” to our ears, which makes it hard to judge a limiter honestly. Unity gain removes the Gain boost from what you hear, so you compare the sound at the same loudness. If the mix sounds worse with the limiter at equal loudness, you are limiting too hard or using the wrong style. The meters still show the real output level. Turn it off before exporting.

Delta

Delta plays only what the limiter is taking away: the original signal minus the limited signal. It is the most honest way to hear the limiter working.

What you hear in Delta	What it means
Silence	The limiter is not doing anything.
Short clicks or ticks	Only peaks are being caught. Usually fine and inaudible in the mix.
Recognisable kicks, snares, bass or vocals	The limiter is eating into the punch and body. Lower Gain or try a gentler style (Transparent, Safe, Dynamic).
Crunch or distortion	A clipping or colour style is adding grit. Decide if it is character or too much.

Why you can hear kicks and bass with Gain at 0: if your mix already peaks above the ceiling (-1 dB by default), the limiter is working even without extra gain. Soft-knee styles (Bus Glue, Vari-Mu Tube, Opto) also start reducing a few dB below the ceiling, and clipping styles round the loudest peaks. Kicks and bass are usually the loudest parts, so they are touched first. To check, choose Transparent, set the ceiling to 0 dB: if the GR meter stays at 0, Delta is silent.

While Delta is on, the final safety clip and dither are switched off, because the difference signal is not meant to be your output. Always turn Delta off before exporting.

10. Meters and loudness

Meter	What it shows
L / R	Output sample peak level from -48 to 0 dBFS, with the peak value in cyan below.
GR	Gain reduction from 0 to 24 dB, growing downwards, with the value in pink below.
Momentary	Loudness over the last 0.4 seconds (EBU R128).
Short-term	Loudness over the last 3 seconds. Useful for comparing sections of a song.
Integrated	Average loudness since the last reset, with the standard EBU R128 gating that ignores silence and very quiet parts. This is the number streaming services use.

- Press **Reset loudness** and play the whole song from start to finish to get a correct Integrated value.
- Loudness is measured on the real output, before Unity gain, so the readings stay correct while you monitor at equal loudness.
- The loudness meter follows the ITU-R BS.1770 standard, including K-weighting.

11. Latency

Lookahead and oversampling delay the audio slightly. Galaxy Limiter reports this delay to your DAW, which compensates automatically, so everything stays in time.

Setting	Added latency at 48 kHz (approx.)
Lookahead 3 ms, oversampling off	3 ms (144 samples)
Lookahead 3 ms, 4x oversampling	about 3.5 ms
Lookahead 10 ms, 16x oversampling	about 11 ms

Changing Lookahead, Oversampling or True peak changes the latency and resets the limiter for a moment. Some DAWs only pick up the new latency after you stop and start playback. For recording through the plugin, use Lookahead 0 and Oversampling off.

12. Troubleshooting

Problem	Solution
zsh: event not found	You pasted a script's text into Terminal. Paste only the commands from section 2.
“build is not a directory”	The set-up step failed. Run `rm -rf build` then `cmake -B build -DCMAKE_BUILD_TYPE=Release` and read its last lines for errors.
cmake: command not found	Install CMake: `brew install cmake`.
Plugin doesn't appear in Logic	Run `killall -9 AudioComponentRegistrar`, restart Logic, and check the Plug-in Manager.
Plugin doesn't appear in Cubase, LUNA or Fender Studio Pro	Rescan plugins in the DAW's plugin manager. Check the file is in the VST3 folder.
Old version still loads	Quit the DAW completely, rebuild, and reopen it.
Distortion on loud parts	Lower Gain, lengthen Attack or Release, raise Oversampling, or choose Transparent or Safe.
Pumping	Lengthen Release or choose Dynamic, Bus Glue or Opto.
Stereo image moves	Set Stereo link to 100 %.
Output quieter than expected	Check Unity gain is off.
Strange thin sound	Check Delta is off.
High CPU	Lower Oversampling. 16x is very demanding.

13. Specifications

Item	Detail
Formats	VST3, AU (macOS), Standalone
Channels	Mono and stereo
Sample rates	Any rate supported by the host (44.1 to 192 kHz)
Limiter	Lookahead peak limiter, dual-stage program-dependent release, soft-knee gain computer
Oversampling	2x, 4x, 8x, 16x linear-phase FIR
True peak	At least 4x oversampling with 0.1 dB safety margin
Loudness metering	ITU-R BS.1770 / EBU R128: momentary, short-term, gated integrated
Dither	TPDF, 16 or 24 bit
Framework	JUCE 8.0.4, C++17, CMake
Version	1.1.0

14. Backup contents and version history

This backup contains everything needed to rebuild Galaxy Limiter from scratch. Keep a copy in a safe place, such as an external drive or cloud storage.

```
GalaxyLimiter/
  Galaxy_Limiter_User_Manual.pdf   this manual
  README.md                        short build notes
  CMakeLists.txt                   build settings, plugin name and IDs
  Source/
    LimiterStyles.h                the 10 style settings (edit to retune)
    LimiterEngine.h                limiter and clipper sound engine
    LoudnessMeter.h                LUFS loudness meter
    PluginProcessor.h / .cpp       parameters, oversampling, dither, metering
    PluginEditor.h / .cpp          space-themed interface
```

The `build` folder is not included. It is recreated automatically the first time you build.

Version history

Version	Changes
1.1.0	Renamed to Galaxy Limiter. New space-themed interface (starfield, neon knobs, glass panels, animated logo, LED meters). Fixed Delta leaking bass and kick with no limiting. Colour styles now only add harmonics while limiting. Added this user manual.
1.0.0	First version: lookahead limiter, 10 styles, oversampling, true peak, dither, Unity gain, Delta, history display and LUFS metering.